# DEPC-treated water (nuclease-free water for RNA and DNA work)

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v1.0, 2026-10-21; 0.1% DEPC, overnight, autoclaved

**Purpose:** water free of RNases (and DNases) for elution buffers, PCR set-up, RNAlater and library work. DEPC (diethyl pyrocarbonate) inactivates nucleases by carbethoxylating histidine residues; autoclaving then destroys the remaining DEPC (it decomposes into $CO_2$ and ethanol).

**Reagent:** diethyl pyrocarbonate, molecular biology grade (e.g. HiMedia MB076, himedialabs.com/us/mb076-depc.html). Store the DEPC bottle at 2–8 °C, tightly closed, away from moisture; open it only in the fume hood.

**Safety:** DEPC is volatile, a suspected carcinogen and reacts violently with ammonia and amines. Fume hood, nitrile gloves, lab coat, safety glasses. Never add DEPC to buffers that contain Tris, HEPES or any other amine: treat the water first, then dissolve the buffer salts in the treated, autoclaved water.

## Procedure

### TREATMENT

**1.** In the fume hood, add **1 mL** DEPC to **1 L** Milli-Q (or double-distilled) water in a clean glass bottle (final **0.1% v/v**). Use a bottle with a loose cap or foil; never a sealed vessel.

**2.** Shake or stir vigorously until the DEPC droplets have dispersed (DEPC is poorly soluble), then stir **≥ 1 h** more.

**3.** Incubate **overnight (≥ 12 h) at 37 °C** (room temperature works with a longer incubation). Keep the bottle in the hood or a ventilated incubator: DEPC vapour escapes during this time.

### INACTIVATION

**4.** Autoclave **121 °C, 20 min** with the cap loosened. This hydrolyses the remaining DEPC; the water must smell of nothing afterwards. A sweet, fruity smell means DEPC is still present: autoclave again.

**Note** Un-destroyed DEPC carbethoxylates adenine residues and will inhibit downstream enzymes. Do not skip or shorten the autoclave step.

**5.** Let the bottle cool with the cap loosened, then close it tightly. Label it “DEPC water”, date, initials.

### USE AND STORAGE

**6.** Store at room temperature, tightly closed. Open only with nuclease-free technique (gloves, clean bench, never pipette back into the bottle). Aliquot into 50 mL nuclease-free tubes for daily use and discard an aliquot once it has been opened for more than a month.

**7.** Buffers: dissolve the salts in DEPC-treated, autoclaved water and autoclave the buffer again. Do not treat the finished buffer with DEPC.

**Note** Commercial nuclease-free water is an acceptable substitute for small volumes; DEPC water is the lab's stock for elution, SD-buffer dilution and RNAlater preparation.

### Materials

| # | Material |
|---|---|
| 1 | Diethyl pyrocarbonate, molecular biology grade (HiMedia MB076 or equivalent) |
| 2 | Milli-Q or double-distilled water |
| 3 | 1 L glass media bottle (autoclavable) with loosened cap or foil |
| 4 | Magnetic stirrer and bar (or shaker) |
| 5 | **37 °C incubator; autoclave** |
| 6 | Fume hood, nitrile gloves, safety glasses |